exome_Scripts: contents

Folder: ~/Desktop/whole_exome_sequencing/exome_Scripts | Updated 17 Oct 2026

ClawBio exome skills, plus the helper scripts and reference data needed to analyse an exome from a VCF. Everything was **copied**; the original analysis folders are unchanged. Run: **run_exome.sh <vcf> <output folder>**. Works offline; no patient data is sent anywhere.

1. ClawBio skills (clawbio_skills/)

Skill folder	Main script	Size	Used by run_exome.sh?
clinical-variant-reporter	clinical_variant_reporter.py, acmg_engine.py, self_audit.py	165 KB	Yes, through the local copy in acmg_local/. The original sends variants to the Ensembl web service.
rare-high-impact-variants	rare_high_impact_variants.py	36 KB	Yes (offline): rare loss-of-function count.
variant-annotation	variant_annotation.py	81 KB	No: uses the Ensembl web service; replaced by local VEP.
vcf-annotator	vcf_annotator.py	47 KB	No: annotates over the web; replaced by local VEP.
wes-clinical-report-en	wes_clinical_report_en.py	76 KB	No: needs a Novogene-style markdown input; marked PROPRIETARY in its metadata.
clinical-variant-prioritizer	prioritize.py, api.py	16 KB	No: built for array or genome genotype lists (rsID input).

Copied from the installed ClawBio package (version 0.7.1) in the exome conda environment. Cache folders were left out.

2. Runner and helper scripts

File	Origin	Size	What it does
run_exome.sh	Written from the sample-2 pipeline	3 KB	The only command you need: run_exome.sh <vcf> <output folder>. Filter to targets, ClinVar, VEP (offline), ACMG, rare loss-of-function count, ClinVar table. Works with filtered and unfiltered DNAscope VCFs.
acmg_local/	ClawBio clinical-variant-reporter, adapted	97 KB	ACMG/AMP 28-criteria classifier reading local evidence instead of the Ensembl web service. Rules unchanged.
helpers/parse_vep.py	Copied from analysis/	3 KB	Flattens VEP output to vep_table.tsv and prepares the rare-variant skill input.
helpers/build_local_evidence.py	Copied from analysis/	3 KB	Builds ACMG evidence from local VEP and ClinVar.
README.md	-	3 KB	How to run, outputs, how to update ClinVar.
environment.yml	conda env export of 'exome'	12 KB	Exact package versions, to rebuild the environment elsewhere.
make_contents_pdf.py	-	7 KB	Builds this PDF.

3. Reference data

File	Origin	Size	What it is
reference/targets.bed	analysis/targets.bed	7 MB	Exome capture targets (3 columns from hg38_exome_comp_spikein_v2.0.2).
reference/clinvar_chr.vcf.gz (+ .tbi)	analysis/	190 MB	ClinVar GRCh38, release 29 Sep 2026, chr-prefixed.
reference/chrmap.txt	analysis/	202 B	Chromosome renaming map, for updating ClinVar.

4. Removed on request

Item	Note
original_runs/ (exact sample-2 script)	Sample-specific record; the original is still in analysis2/run_sample.sh.
trio/ (run_trio.sh, trio_qc.py, trio_analysis.py) and hpo/	Built for the prenatal trio, with that family's HPO terms. trio_analysis.py and the HPO files are still in analysis_trio/. run_trio.sh and trio_qc.py existed only here and were deleted.
reference/epi_genes_list.txt, reference/ymsy.bed	Case-specific (epilepsy gene list; trio sex check). Originals still in analysis/ and analysis_trio/.

5. Not included

Item	Reason
VEP cache (~/.vep, about 52 GB)	Too large to duplicate; run_exome.sh uses it in place. README explains how to reinstall.
PDF report generators	They were in a temporary folder that has since been cleared.
Patient results	Stay in the analysis folders; exome_Scripts holds only tools and reference data.

6. Test status

Passed on 17 Oct 2026. run_exome.sh was re-run on sample-2 after the restructure and compared with the original analysis: same counts (63,257 on target, 30,460 variants); vep_table.tsv and the ACMG classification table are byte-for-byte identical (Pathogenic 0, Likely pathogenic 33, VUS 2,294, Likely benign 700, Benign 25,928); the rare loss-of-function result matches.

Total folder size: 197 MB